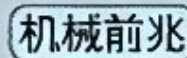
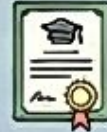


UPSC PRELIMS 2026: ELECTORAL RESET & FEDERALISM

I 1. THE LOK SABHA EXPANSION (LEGAL PACKAGE 2026)

-  **Constitutional Basis:** Constitution (131st Amendment) Bill, 2026 amends Articles 11 &  
-  **The 2011 Census Pivot:** Delimitation decoupled from post-2026 census. Uses 2011 data.
- ★ **Seat Surge:** Increase strength from 543 to 850  →  
- ★ **Women's Reservation:** Precursor to operationalizing 33% quota by 2029.  

III 3. THE HYBRID MODEL: A WAY FORWARD

-  **Balanced formula** balancing “One Person, One Vote” with Federal Fairness. 
- ★ **Allocation Criteria:**  
- **50% Population:** Based on 2011/Current Census.
 - **50% Performance:** Based on State GDP (GSDP), Fiscal Contribution, Literacy/Health.
- ★ **Objective:** States with high exchequer/social indices rewarded political representation.  

II 2. STALIN'S CRITIQUE: “THE SWORD OVER SOUTH”

-  **Tamil Nadu CM M.K. Stalin & Southern leaders** raise Federal Alarm. 
- ★ **Punishment for Success:** Penalized for Population Control & high HDI. 
- ★ **Relative Power Dilution:** South share shrinks ~24% to ~19%. Voice silenced. 
- ★ **Demand:** Maintain 1971 relative weightage or use “pro-rata” increase. 

IV 4. KEY FEDERAL SAFEGUARDS (UPSC POINTERS)

-  **Delimitation Bill, 2026** shifts “population” definition. 
-  “population as determined by Parliament by law”.
- ★ Grants significant political flexibility for redistributions.



ABHISHA AI by Abhishek Pathak

UPSC PRELIMS 2026: INDIA'S VULNERABILITY TRAP

I CONTEXT: WEALTH CONCENTRATION & THE SQUEEZE



- ★ **Macroeconomic Disconnect:** GDP growth (bar chart) vs Stagnant real well-being (flat line).
- ★ **Wealth concentration at Top** rising.
- ★ **Middle Class Vulnerability definition** (risk meter): Prob. of non-poor household falling below line due to risks (Health shock, Job loss, Inflation).



- ★ Limited creation of high-productivity formal jobs.

III ANALYTICAL FRAMEWORK: BEYOND INCOME (THE WB APPROACH)

A. THE WORLD BANK WELL-BEING SPECTRUM



- ★ Spectrum moves beyond binary metrics.
- ★ **India Challenge:** policy successful from 'Extreme' to 'Vulnerable', now need to move from 'Vulnerable' to 'Secure Middle Class'.

II CHALLENGE: POVERTY SHIFT & THE PRECARIETY TRAP



New Segment:
Households just above Extreme Line are ~30% (vs earlier 50%).

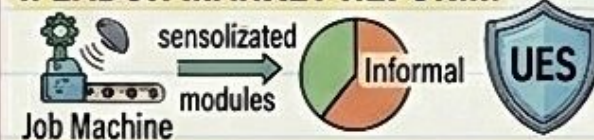


Extreme Poverty Rate: fallen (using \$2.15/day line).
Status: "Vulnerable," highly susceptible to ~30%.

- ★ **Precarity Trap**
Not in "extreme poverty," but no upward mobility.
- ★ Status: "Vulnerable," highly to falling back if crisis occurs.

IV THE WAY FORWARD: TO SECURITY & SHARED PROSPERITY

1. LABOR MARKET REFORM:

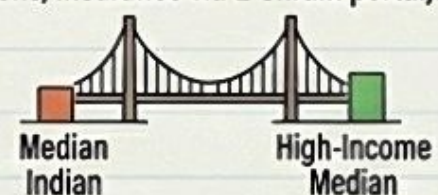


2. DE-RISKING HOUSEHOLDS:

Drastically reduce Out-of-Pocket Pensions, Insurance via E-Shram portal).

3. MSME ECOSYSTEM INTEGRATION:

Build Micro, Small, Medium Enterprise capacity (MSMEs create jobs). support with credit, technology, global value chains.



4. GLOBAL PROSPERITY GAP:

NITI MPM Score Falling (Refined focus on reducing precarity).

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UPSC CSE 2026: BALANCING DIGITAL ECONOMY & COSMOLOGY

DIGITAL LANDSCAPE (2026): REGULATORY BACKING & VALUE CLASSIFICATION

PART 1: DIGITAL ASSETS & BLOCKCHAIN (S&T)

1. DEPOSIT TOKENS



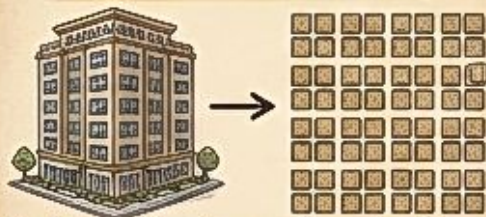
Concept: "Tokenized" form of bank account money. Unlike private crypto, these are regulated liabilities of a bank.
Backing: Traditional fiat (₹) in a regulated bank.
Example: Banks (e.g., JPM Coin) for instant client transfers.

2. STABLE TOKENS (STABLECOINS)



Cryptocurrencies designed to have a stable price. Used as medium of exchange to avoid volatility.
Backing: Usually 1:1 by fiat (USD/INR), gold, or algorithm.
Example: USDT (Tether) = \$1 USD.

3. ASSET TOKENS (RWA - TOKENIZED REAL-WORLD ASSETS)



Real Estate/Bonds → Property Tokens
Concept: "Fractional Ownership" (e.g., 1% of a building). **Backing:** Real-world assets.
Example: Rent share from fractional apartment ownership.

4. NON-FUNGIBLE TOKENS (NFTs)



Unique digital ownership certificates. Cannot be exchanged 1:1. Prove "Digital Scarcity".
Backing: Digital provenance (ownership history) on the blockchain.
Example: Digital Art (e.g., Beeple) or Metaverse Land.

PART 2: COSMOLOGY & THE EXPANDING UNIVERSE (GEOGRAPHY/PHYSICS)

"CRISIS IN COSMOLOGY": HUBBLE TENSION & NEW PHYSICS

1. MEGAPARSEC (Mpc)



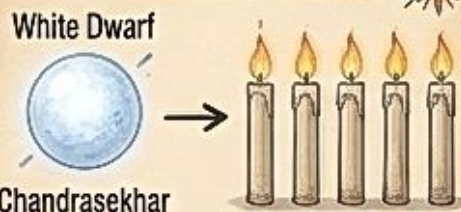
Unit of distance in extragalactic astronomy.
 $1 \text{ Megaparsec} = 10^6 \text{ parsecs} \approx 3.26 \text{ million light-years}$.
Importance: Denominator for universe expansion rate.

2. CEPHEID VARIABLE STARS



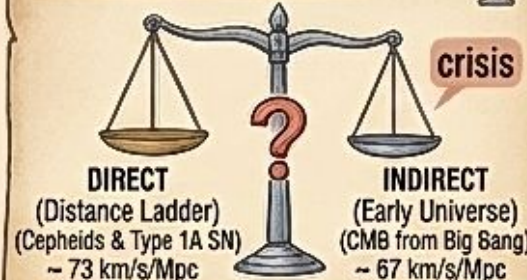
"Cosmic Yardsticks" for shorter distances.
Concept: Pulse rate (period) is directly related to true brightness.
Use Case: Find distance by comparing True vs. Apparent brightness.

3. TYPE 1A SUPERNOVA



White Dwarf → Standard Candles
Chandrasekhar Limit $1.44 M_{\odot}$ → Standard Candles
"Standard Candles" for long distances.
Concept: Occurs when a White Dwarf star explodes at a specific mass limit.
Use Case: Uniform brightness acts as a distance beacon.

4. HUBBLE TENSION

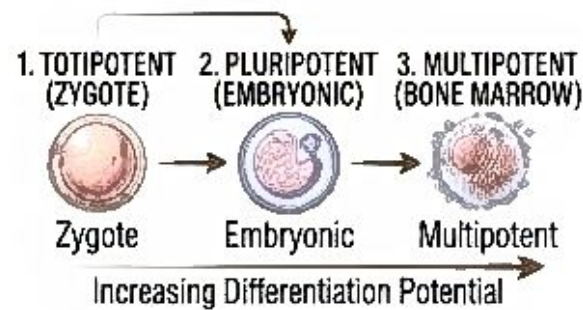


DIRECT (Distance Ladder) (Cepheids & Type 1A SN) ~ 73 km/s/Mpc
INDIRECT (Early Universe) (CMB from Big Bang) ~ 67 km/s/Mpc
Major Discrepancy in Hubble Constant (H_0).
Crisis: The 6 km/s/Mpc gap suggests missing Standard Model physics (e.g., Dark Energy, exotic particles).

UPSC CSE 2026: THE COMPREHENSIVE ECO-BIO CONCEPT REVISER

SECTION 1: STEM CELL TECHNOLOGIES & THE BIO-INNOVATION (S&T)

Progression 1: Understanding Stem Cell Potency



The Concept (Potency): A hierarchy of cell states, from maximum potential (Totipotent) to specific roles.

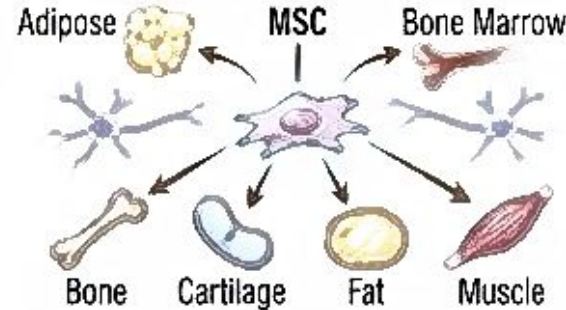
Signaling Mechanism: "Cellular Paracrine Signaling"—cell releasing small protein "whispers" to neighbors.

Factual Points: Focus on "Chandrasekhar Limit for Potency" metaphor, FDA-approved Leukemia treatments, regenerative medicine.

"Mesenchymal" vs. "Other" Stem Cells (Table)

Potency	Signaling	Harvesting
Multipotent vs. Pluripotent	High Immunomodulation	Non-Invasive
---	vs. variable	Non-Invasive vs. Invasive (Embryonic)

Progression 2: The Role of MSCs (Mesenchymal Stem Cells)

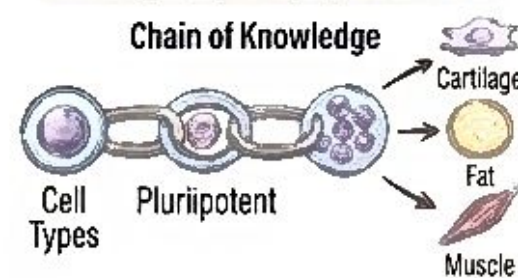


The Concept (Specificity): Key to success. MSCs from Bone Marrow & Adipose differentiate into specific tissues.

Signaling Mechanism: "Immunomodulatory Signaling"—MSC releasing "shield" cytokines to pacify an inflammatory immune cell (transplant rejection prevention).

Factual Points: "National Guidelines 2026" (regulatory context); sources: bone marrow, fat.

KEY CONCEPTS OF PART 1



SECTION 2: ECOLOGICAL SPECIES' ROLES & CONSERVATION

Progression 3: Keystone Species & Ecosystem Foundation

Keystone (e.g., Sea Otter)



The Concept (Foundation): Classical arch, Sea Otter is literal keystone. Removing it collapses Kelp Forest ecosystem. Shows "Low Biomass, High Impact".

Signaling Mechanism: "Trophic Cascade Signaling"—arrow system: Less Otter → More Urchins → Less Kelp. Population control.

Factual Points: Examples: "Apex Predators" and "Ecosystem Engineers" (e.g., Elephants creating water holes).

"Keystone" vs. "Sentinel" (Table)

Primary Value	Signal Type
Structural Stability vs. Population Control	Early Warning Early Stress Detection

Progression 4: Sentinel Species & Sensory Alerts

Alert (e.g., Canary in a Coal Mine)

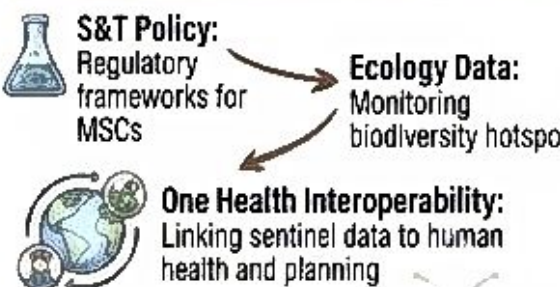


The Concept (Alert): Canary in mine provides "alerts" of danger (CO gas).

Signaling Mechanism: "Sensory Signaling"—a sensitivity-time graph showing Sentinel reacts sooner to toxin.

Factual Points: Examples for Air Quality, Frogs for Water Pollution, Vultures.

SECTION 3: STRATEGIC ACTIONS & INTEROPERABILITY ROADMAP (One Health)



STRATEGIC ROADMAP FOR NATIONAL COOKSTOVE INITIATIVE & CPCB STANDARDS

Part 1: NATIONAL STRATEGY FOR SUSTAINABLE BIOMASS (GS-3)

1. BIOMASS STOVER: RESOURCE & CHALLENGES



***Note:** "Stover" is defined as stalks and leaves.

2. MODERN FORCED DRAFT IMPROVED COOKSTOVE (GS-3)

Mechanism: Optimized gasification for total fuel conversion.

Outcome: Significant reduction in harmful indoor air pollutants.

2. MODERN FORCED DRAFT IMPROVED COOKSTOVE (GS-3)

1. Energy Potential: Essential for Next-Gen Biofuels.

2. Challenges: Open burning causes IAP & Black Carbon emissions.



BIOMASS COOKSTOVE COST ANALYSIS (FORCED DRAFT)

Cost Factor	Traditional Chulha	Modern ICS (e.g., Forced Draft)
Initial Capital	₹0 (virtually free)	₹2,000-₹4,000 (High initial Cost)
Operating Cost	₹0	Variable (e.g., pellets/processing)
Subsidy Dependence	Low	High (National Cookstove Initiative)



****INTEGRATED LIFE SCHEME FOR RURAL ADOPTION**

- Combining health, finance, and skill programs.

Part 2: ECOLOGICAL GUIDELINES FOR RITUAL OFFERINGS AND IMMERSIONS

"Crisis in Rituals": Environmental concerns & framework



Offerings: Flowers, idols, food, milk, oil.

The Concept: Rituals often have large ecological footprints.

Impacts: POP/lead paints → heavy metals; organic offerings → eutrophication.

Ritual Offerings vs. Sustainability Table

Offering Type	Env. Consequence	Mitigation/Sustainable Alt.
PoP Idols	Heavy Metal Poisoning	Seed Idols (grow into plants)
Organic Matter	Eutrophication	Artificial Folds for decomposition
Non-biodeg Items	Drainage clogging	Strict non-use/tines

***Note:** Reducing IAP is reserved for special cases where stalks and leaves cannot be treated.

4. REGULATORY FRAMEWORK & LEGAL PRINCIPLES

National Green Tribunal (NGT) Directives heavy fines for non-compliance.

CPCB Guidelines (2020): PaX Idol ban and artificial ponds requirement.



Judicial Precedents: Polluter Pays Principle (M.C. Mehta vs. UOI). (Update: Landmark UOI case established the principle).

Legal Basis: Water Act 1974, Article 51A(g) 51A(g) & 25 **Conflict:** Cultural Rights vs. Env. Duties.

INTEROP STANDARD DEVELOPMENT & POLICY INTEGRATION



Regulatory Alignment: NGT & CPCB Standard Harmonization

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SPECIALIZED LEGAL TOPICS: HEAT WAVES & PIRACY LEGAL FRAMEWORKS

Part 3: HEAT WAVES & THERMAL INJUSTICE (GS-2/GS-3)

The 'Right to Cool' Framework

Constitutional Basis: Argument to recognize the Right to Cool under Article 21 (Right to Life and Dignity).



Structural Reforms:

- **Heat Action Plans (HAPs):** Moving beyond emergency response to urban design (cool roofs, urban forests).
- **Labor Laws:** Adjusting working hours for outdoor workers during peak heat.



The Legal Gap:

Currently, a Heatwave is NOT classified as a 'Natural Disaster' under the Disaster Management Act, 2005. This limits the release of central relief funds (SDRF/NDRF) compared to floods or cyclones.



Part 4: LEGAL CONSEQUENCES OF PIRACY (GS-2/GS-4)

Case Study: Leak of 'Jana Nayagana'

Issue: Digital piracy leads to massive revenue loss and infringes on moral/economic rights of creators.



Primary Legal Framework:

Copyright Act, 1957: Protects original literary, dramatic, musical, and artistic works.



Cinematograph Act, 1952: Recently amended to include strict penal provisions for unauthorized recording and exhibition.

Key Sanctions:

Criminal: Imprisonment (3 months to 3 years) and heavy fines (up to to 5% of production cost).



Civil: Injunctions (John Doe orders) to block websites/links before they even leak content.

Ethical Dimension:

Piracy as a violation of intellectual property and the 'sweat of the brow' doctrine.



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